

Dynamic Anchor-Based One-Step Hypergraph Ensemble Clustering

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Abstract—Ensemble clustering aims to derive a more robust consensus result from a set of base clustering results. Recently, anchor-based clustering methods have improved computational efficiency by learning the relationships between anchors and samples, thereby avoiding the expensive computation of pairwise sample similarities. However, these methods typically rely on a static anchor selection strategy and require post-processing to obtain the final clustering assignments. This often leads to inflexible anchors and a low-quality anchor similarity matrix, ultimately degrading clustering accuracy. To address this issue, we propose a novel ensemble clustering method named Dynamic anchor-based One-step hypergraph Ensemble Clustering (DREAM). Specifically, DREAM first transforms the base clustering results into a hypergraph. It then introduces a novel hypergraph enhancement strategy to improve the reliability of the initial hypergraph. Next, DREAM introduces a mapping matrix to filter redundant information in the hypergraph, and reconstructs the hypergraph via matrix factorization to obtain the anchor similarity matrix. Subsequently, DREAM introduces an alignment mechanism that refines the anchor similarity matrix by generating local consensus information. The alignment is achieved by maximizing the element-wise consistency between the anchor similarity matrix and the local consensus information. This yields a high-quality anchor similarity matrix that can be directly projected into the label space, enabling one-step generation of clustering results without any additional post-processing. Extensive experimental results demonstrate the superior performance of the proposed DREAM method.

Index Terms—One-step ensemble clustering, dynamic anchor learning, hypergraph reconstruction.

I. INTRODUCTION

ENSEMBLE clustering integrates multiple base clustering results to obtain a robust consensus partition [1], [2], [3]. Since it operates solely on decision-level outputs, i.e., base clustering results, without accessing original features, it inherently preserves feature-level data privacy [4], though it may limit the

exploitable information [5]. For instance, in privacy-sensitive scenarios such as cross-institutional medical analysis, financial risk profiling, and customer segmentation, raw feature vectors may be inaccessible due to data protection regulations, proprietary restrictions, or high transmission costs. In such cases, different parties can provide only base clustering results while keeping the original features undisclosed.

Most existing ensemble methods convert base results into a co-association (CA) matrix, since the CA matrix records how frequently each pair of samples falls into the same cluster and thus captures clustering structure at the sample level. However, noise in the base clusterings often degrades the initial CA matrix. To address this, researchers often enhance CA quality using various learning or graph-based techniques, followed by hierarchical or spectral clustering to derive the final discrete consensus result [6], [7]. Unfortunately, the $n \times n$ size of the CA matrix leads to $O(n^2)$ or even $O(n^3)$ time complexity, hindering scalability to large datasets.

To address this scalability issue, anchor-based ensemble clustering methods have been proposed. These methods integrate multiple base clustering results in three stages: (i) selecting representative samples (anchors), (ii) learning the anchor-sample similarities (i.e., the anchor similarity matrix), and (iii) performing clustering, without constructing the full $n \times n$ co-association matrix [8], [9]. Anchor selection is crucial because it determines the quality of the anchor similarity matrix, which directly impacts clustering accuracy. Existing methods typically use cluster centroids from K-means or its variants (e.g., BKHK [10]) as anchors, which remain fixed once selected [8], [11], [12]. This static strategy severely limits the flexibility and adaptability of learning similarities between anchors and samples, resulting in a low-quality anchor similarity matrix. In practice, achieving satisfactory performance often requires a large number of anchors, which further escalates the computational cost. Moreover, since existing methods rely on raw data features for anchor selection, they must access feature-level inputs in addition to the base clustering results, thereby undermining the inherent privacy-preserving advantage of ensemble clustering. Taken together, these limitations highlight the need for a feature-free, one-step dynamic anchor learning method. This method should operate solely on base clustering results to preserve privacy and avoid post-processing steps (e.g., applying k -means to the anchor similarity matrix) when obtaining the final clustering labels. At the same time, it should dynamically learn high-quality anchors and their similarities to samples, thereby enhancing both flexibility and accuracy. Specifically, such a

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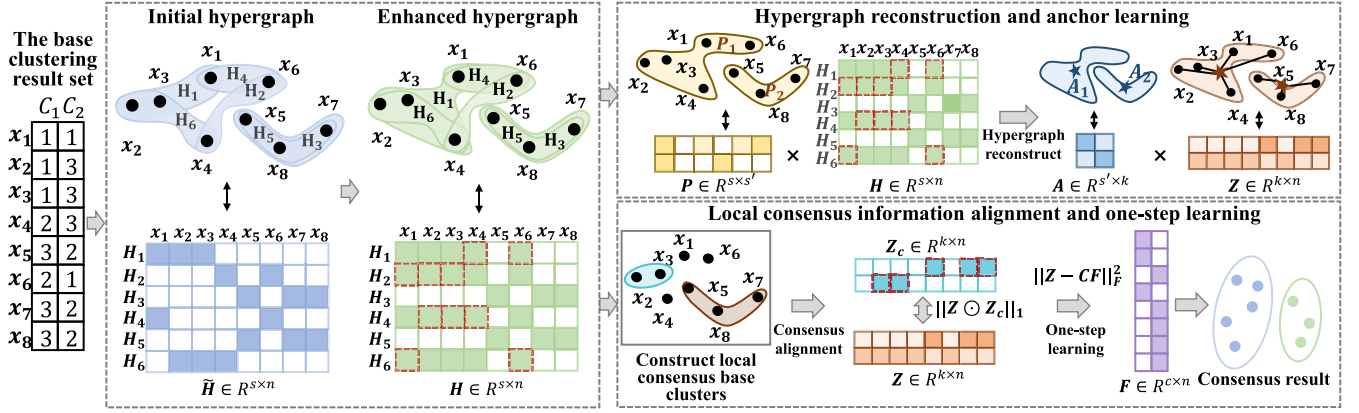


Fig. 1. The framework of the proposed DREAM method. DREAM first converts multiple base clustering results into a hypergraph and enhances it via high-order random walks over hyperedges. Then, dynamic anchors are learned to reconstruct the enhanced hypergraph and obtain the anchor similarity matrix. Local consensus information is further constructed from base clusters to guide similarity alignment. Finally, DREAM directly learns the consensus partition in a one-step manner without post-processing.

method should address the following challenges. (C1): How to improve the reliability of base clustering results to enhance the accuracy of learning anchor-sample similarities. (C2): How to dynamically select anchors and construct a high-quality anchor similarity matrix using only base clustering results. (C3): How to exploit the local consensus information embedded in the base clustering results within a one-step ensemble clustering paradigm to improve clustering accuracy.

To address these challenges, we propose a novel ensemble clustering method named Dynamic Anchor-based one-step Hypergraph Ensemble Clustering (DREAM). As illustrated in Fig. 1, DREAM first converts the base clustering results into a hypergraph and employs random walks to capture high-order similarities among hyperedges. Then, DREAM adjusts the hypergraph structure according to these high-order similarities, producing a more stable and reliable enhanced hypergraph. Next, DREAM introduces a mapping matrix to filter out redundant information and reconstructs the hypergraph via matrix factorization. During reconstruction, both the anchor matrix and the anchor similarity matrix are incorporated into the optimization objective, enabling dynamic learning. Subsequently, DREAM introduces a local consensus information alignment mechanism, which aligns the anchor similarity matrix with the local consensus information by maximizing their element-wise consistency. This alignment strategy encourages the anchor similarity matrix to preserve the underlying local consensus information and further improves its quality. Based on this high-quality anchor similarity matrix, DREAM can then directly project it into a label space to obtain the final consensus clustering results, without any additional post-processing steps.

The main contributions of this paper are:

- We propose a novel ensemble clustering method named dynamic anchor-based one-step hypergraph ensemble clustering (DREAM). Compared to CA matrix-based methods, DREAM reduces the computational complexity, making it more suitable for large-scale datasets.
- We devise a hypergraph enhancement strategy to improve the reliability of the initial hypergraph and introduce an

alignment mechanism to ensure that local consensus information is preserved.

- We design an alternating optimization strategy for the objective function and demonstrate DREAM's effectiveness by comparing it with 17 representative clustering methods.

Discussion. This paper substantially extends our earlier conference version, YACHT [13], which appeared in the Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI-25). The main technical difference is that DREAM reformulates the previous two-stage framework into a one-step reconstruction-and-assignment model. Specifically, DREAM directly introduces the discrete assignment matrix $\mathbf{F}$ into the optimization objective through $\|\mathbf{Z} - \mathbf{CF}\|_F^2$, so that the final consensus labels can be obtained from $\mathbf{F}$ without additional post-processing. This design allows the model to make more effective use of the information contained in the anchor similarity matrix and avoids the information degradation caused by a separate post-processing stage.

In addition, DREAM designs a local consensus information alignment mechanism that operates directly on the anchor similarity matrix $\mathbf{Z}$. In the previous conference version, the alignment was restricted to relating the local consensus information to the label partition generated by the hypergraph regularizer. By contrast, aligning $\mathbf{Z}$ itself removes an intermediate label-space feedback stage and further alleviates information loss. These methodological changes are reflected in the optimization objective and algorithmic design, and are validated by additional ablation and comparison experiments. Finally, we evaluate DREAM on substantially larger-scale datasets and provide more comprehensive parameter studies and additional evaluation metrics.

II. RELATED WORK

A. Ensemble Clustering

Ensemble clustering is an effective technique for improving the applicability of clustering algorithms across data with diverse distributions and modalities [14], [15]. The objective of ensemble clustering is to extract sample similarity relationships from

base clustering results generated by diverse base clustering algorithms, thereby deriving a robust consensus result. This decision-level (base clustering results) fusion strategy imparts feature-level privacy protection to ensemble clustering by rendering the sample features inaccessible [16], [17]. Although some methods incorporate raw feature data for richer information [18], [19], this compromises the inherent privacy advantage of ensemble clustering. Therefore, the vast majority of ensemble clustering methods rely solely on the input of base clustering results to obtain consensus, making the quality of these base clustering results (ensemble members) critically important. Existing ensemble clustering methods can be categorized into methods based on ensemble selection and methods based on fusion strategies. Methods based on ensemble selection aim to enhance clustering accuracy by balancing the two key metrics that govern ensemble performance: the accuracy and the diversity of the base clustering results [20], [21]. For instance, Shi et al. proposed TCE-TCES to trade off accuracy and diversity for more robust consensus [22]. Yang et al. formulated these two key metrics as a combined optimization problem to address the issue [23]. Methods based on fusion strategies focus on designing robust and effective fusion techniques to enhance the accuracy of the consensus result, striving to obtain robust results from multiple weak base clustering results [24]. Mainstream fusion strategies construct a CA matrix to mine sample similarities [25], [26]. For example, Zhou et al. integrated self-paced learning [27], while Jia et al. and Huang et al. incorporated tensor learning [1], [6]. Bian et al. proposed a weighted adaptive ensemble method using a fuzzy CA matrix [4], and Xu et al. enhanced the CA matrix with high-order neighborhood information [7]. Qin et al. explored the shared latent space of multiple CA matrices for ensemble clustering [28]. Zheng et al. introduced a method based on structured graph learning [29]. Cao et al. proposed spectral ensemble clustering via tensor graph learning [30]. Wei et al. incorporated self-supervised learning into a self-constrained ensemble method [31]. However, CA matrix-based methods face a major computational efficiency bottleneck (time complexity $O(n^2)$), limiting scalability.

Some methods represent the membership relationships between samples and base clusters by transforming base clustering results into a bipartite graph (or hypergraph) form, achieving ensemble clustering through learning these membership relationships. For instance, Zhou et al. used structured hypergraph learning [32], Li et al. formulated ensemble clustering as a k-HyperEdge Medoid discovery problem [33], and Zhang et al. employed structured bipartite graph learning [34].

B. Anchor-Based Clustering

Anchor-based clustering replaces pairwise sample similarities with compact anchor-sample similarities, significantly improving efficiency [35], [36]. It has been widely applied in multi-view and subspace clustering [37], [38], [39], [40]. Recent studies further show that compact representative structures can be collaboratively learned or fused across views/granularities to preserve clustering consistency in fuzzy, multi-view, and incomplete multi-view settings [41], [42], [43]. A key challenge is the

selection of high-quality anchors. The goal is to ensure that the compact anchor similarity matrix can accurately approximate the correct clustering structure embedded in the full sample similarity matrix. For example, Zhang et al. used cross-view representations for anchor alignment [44], and Wang et al. jointly optimized anchor graph construction and alignment [45]. In ensemble clustering, anchors are often obtained through random sampling or by using K-means and its variants. For example, Huang et al. selected anchors using a hybrid strategy combining random selection and the K-means method [11]. Yang et al. acquired anchors via random sampling and then performed spectral clustering [46]. Liang et al. used K-means to obtain anchors and construct a membership matrix [47]. Zhang et al. employed BKHK (a K-means variant) for anchor selection and similarity matrix construction [8]. Although these methods aim for efficiency, the anchors are fixed once chosen, limiting flexibility and accuracy. Moreover, some methods rely on original data features for anchor selection. This compromises the feature-level privacy preservation inherent to ensemble clustering.

Unlike existing methods, we propose a dynamic anchor-based ensemble clustering method that operates solely on base clustering results, without accessing raw features. Our method dynamically updates anchors during optimization. Although this incurs additional runtime, it enables more flexible anchor learning and achieves superior accuracy compared to current fast ensemble methods.

III. THE PROPOSED METHOD

A. Preliminary

Given a dataset $\mathcal{D} = \{x_1, \dots, x_n\}$ and m base clustering results $\Pi = \{C^1, \dots, C^m\}$, where x_i represents the i -th sample. Our goal is to fuse the m base clusterings into a consensus clustering result. For each base clustering result $C^j = \{\pi_1^j, \pi_2^j, \dots, \pi_{s_j}^j\}$, it partitions the n samples into s_j base clusters, where π_h^j represents the h -th base cluster in C^j . From the m base clustering results, we construct a hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \tilde{\mathbf{W}})$, where $\mathcal{V}$ is the node set (each node corresponds to a sample). Each hyperedge in the hyperedge set $\mathcal{E}$ corresponds to a base cluster, yielding a total of $s = \sum_{j=1}^m s_j$ hyperedges. The initial hypergraph indicator matrix $\tilde{\mathbf{H}}$ is constructed as shown in Fig. 1. The main notations are summarized in Table 1.

B. Enhanced Hypergraph and Anchor-Based Reconstruction

To enhance the reliability of the initial hypergraph [32], we introduce a random walk to capture high-order proximities among hyperedges. The direct similarity between hyperedges is computed using Jaccard similarity [48]:

$$\mathbf{O}_{ij} = \text{Jaccard}(\pi_i, \pi_j), \quad \text{Jaccard}(\pi_i, \pi_j) = \frac{|\pi_i \cap \pi_j|}{|\pi_i \cup \pi_j|},$$

where $\mathbf{O}_{ij}$ represents the similarity between hyperedges i and j . By normalizing $\mathbf{O}$ as $\mathbf{O}_p = \mathbb{D}_o^{-1} \mathbf{O}$, it can be interpreted as a probability transition matrix, where $\mathbb{D}_o$ is the degree matrix of $\mathbf{O}$. Then, the high-order relationship matrix $\tilde{\mathbf{O}} \in \mathbb{R}^{s \times s}$ of the

hyperedges is defined as:

$$\tilde{\mathbf{O}} = \mathbf{O}_p^{(1)\top} \mathbf{O}_p + \mathbf{O}_p^{(2)\top} \mathbf{O}_p + \dots + \mathbf{O}_p^{(d)\top} \mathbf{O}_p, \quad (1)$$

where $\mathbf{O}_p^{(d)} = \mathbf{O}_p^{(d-1)\top} \mathbf{O}_p^{(1)}$. Based on this similarity, the initial hypergraph structure can be enhanced. The resulting enhanced hypergraph $\mathbf{H} \in \mathbb{R}^{s \times n}$ is constructed as follows:

$$\mathbf{H}_{i:} = \begin{cases} \Phi \left(\sum_{j \in \{j | \tilde{\mathbf{O}}_{ij} \geq \alpha\}} \tilde{\mathbf{H}}_{j:} \right) & \text{if } \sum_{j \in \{j | \tilde{\mathbf{O}}_{ij} \geq \alpha\}} \tilde{\mathbf{H}}_{j:} > 0, \\ \tilde{\mathbf{H}}_{i:} & \text{otherwise,} \end{cases} \quad (2)$$

where $\Phi(\cdot)$ is a mapping function that converts all positive values to 1, and α denotes the similarity threshold. α controls the fusion threshold of hyperedges, such that those with low similarity are not merged. Although some existing ensemble clustering methods also adopt random-walk strategies over base cluster relations, their propagated similarities are usually directly used for consensus graph construction, meta-cluster discovery, or final graph partitioning. In contrast, our random-walk-based enhancement is not designed to directly produce the final clustering result. Instead, it uses the propagated high-order similarities to identify reliable relations among hyperedges and accordingly adjust the hypergraph structure. The resulting enhanced hypergraph provides a more stable structural representation for the subsequent dynamic anchor reconstruction and one-step learning.

After obtaining the enhanced hypergraph $\mathbf{H}$, a self-expressive model is used to capture sample affinities: $\min_{\mathbf{S}} \|\mathbf{H} - \mathbf{D}\mathbf{S}\|_F^2$, where $\mathbf{S} \in \mathbb{R}^{n \times n}$ is the similarity matrix and $\mathbf{D}$ is the dictionary matrix. However, this formulation has two limitations: (i) the similarity matrix $\mathbf{S}$ is of size $n \times n$, making the reconstruction of $\mathbf{H}$ computationally slow; (ii) it cannot filter out the redundant information in the hypergraph. To address these problems, we introduce anchors and a mapping matrix. The anchors reduce the similarity matrix dimension from $n \times n$ to $k \times n$, where k is the number of anchor points, resulting in a smaller anchor similarity matrix $\mathbf{Z}$ and accelerating the reconstruction. The mapping matrix further helps to remove redundant hyperedges, facilitating the capture of a higher-quality anchor similarity matrix. Consequently, we obtain the following formulation:

$$\begin{aligned} & \min_{\mathbf{P}, \mathbf{A}, \mathbf{Z}} \|\mathbf{P}^\top \mathbf{H} - \mathbf{A}\mathbf{Z}\|_F^2 \\ \text{s.t. } & \mathbf{P}\mathbf{P}^\top = \mathbf{I}, \mathbf{A}^\top \mathbf{A} = \mathbf{I}, \mathbf{Z}^\top \mathbf{1} = \mathbf{1}, \mathbf{Z}_{ij} \geq 0, \end{aligned} \quad (3)$$

where $\mathbf{P} \in \mathbb{R}^{s \times s'}$ represents the projection matrix, $\mathbf{Z} \in \mathbb{R}^{k \times n}$ is the anchor similarity matrix, and $\mathbf{A} \in \mathbb{R}^{s' \times k}$ denotes the anchor matrix. s' is the number of representative base clusters. The orthogonal constraints on $\mathbf{P}$ and $\mathbf{A}$ facilitate learning a more accurate anchor similarity matrix $\mathbf{Z}$ [49]. In DREAM, “dynamic anchors” refer to learnable latent representatives rather than statically selected samples or centroids. The anchor matrix $\mathbf{A}$ is jointly optimized with the anchor similarity matrix $\mathbf{Z}$ through enhanced hypergraph reconstruction. Thus, although the enhanced hypergraph is fixed after construction, the anchor representations are dynamically updated during optimization to better fit its structural information. The number of anchors remains fixed, but their latent representations are not. This differs from static

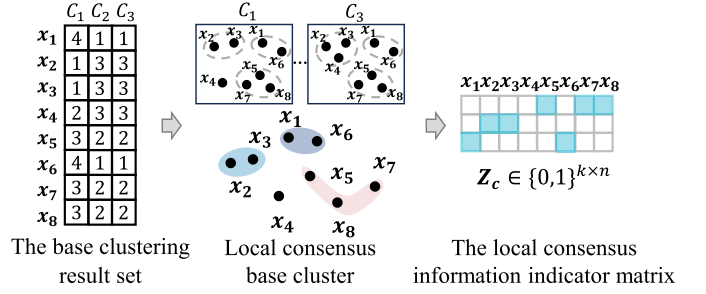


Fig. 2. Samples x_2 and x_3 form a consensus cluster, while $\{x_1, x_6\}$ and $\{x_5, x_7, x_8\}$ also form consensus clusters, respectively. The consensus clusters are ranked according to the number of samples they contain. The top k consensus clusters are selected to generate the matrix $\mathbf{Z}_c$, where k denotes the number of anchors.

anchor selection strategies, where anchors are selected once and kept unchanged throughout optimization.

C. Local Consensus Information Alignment

In ensemble clustering, local consensus information refers to the results on which all base clustering results reach an agreement. For example, two samples are assigned to the same cluster across all base clustering perspectives. Since the CA matrix interprets sample-level similarity, local consensus information can be derived from it. Some methods also adopt this idea [6], [50], [51] to improve clustering accuracy. However, in the context of sample-cluster similarity representations (e.g., hypergraphs), local consensus information has not yet been exploited. Given that the anchor similarity matrix directly impacts clustering accuracy, we therefore enforce the incorporation of local consensus information into the anchor similarity matrix, thereby enhancing its quality. This requires constructing a local consensus indicator matrix with the same dimensionality as the anchor similarity matrix. To this end, we first define a consensus base cluster S .

Definition 1. Let S be a set of samples. The set S is a consensus base cluster if and only if (i) $\forall x_i, x_j \in S, \tilde{\mathbf{H}}_{i:} = \tilde{\mathbf{H}}_{j:}$, and (ii) the number of samples contained in S is at least 2 (i.e., $|S| \geq 2$).

Definition 1 characterizes consensus base clusters derived from the initial hypergraph. Although both consensus base clusters and microclusters [52] require consistent sample-base cluster relationships (i.e. $\tilde{\mathbf{H}}_{i:} = \tilde{\mathbf{H}}_{j:}$), we additionally require that a consensus base cluster contains at least two samples. This is because a microcluster may contain a single sample, but retaining a consensus base cluster with only one sample for alignment is meaningless. Given the limited number of rows in $\mathbf{Z}_c$ (i.e., the number of anchors k), we aim to preserve as much local consensus information as possible from a large number of consensus base clusters. We rank all consensus base clusters by the number of samples they contain, and then retain the top k consensus base clusters. Based on these k consensus base clusters, we construct the local consensus information indicator matrix $\mathbf{Z}_c \in \mathbb{R}^{k \times n}$. Fig. 2 illustrates the construction of the local consensus information indicator matrix $\mathbf{Z}_c$. Note that this

process relies solely on base clustering results, without using the ground truth. We then align $\mathbf{Z}$ with $\mathbf{Z}_c$ by maximizing their element-wise consistency, i.e., $\max_{\mathbf{Z}} \|\mathbf{Z} \odot \mathbf{Z}_c\|_1$, where $\odot$ denotes the Hadamard product. In this way, the anchor similarity matrix is forced to preserve the local consensus information, leading to the following formulation:

$$\begin{aligned} \min_{\mathbf{P}, \mathbf{A}, \mathbf{Z}} \quad & \|\mathbf{P}^\top \mathbf{H} - \mathbf{AZ}\|_F^2 - \|\mathbf{Z} \odot \mathbf{Z}_c\|_1 \\ \text{s.t.} \quad & \mathbf{PP}^\top = \mathbf{I}, \mathbf{A}^\top \mathbf{A} = \mathbf{I}, \mathbf{Z}^\top \mathbf{1} = \mathbf{1}, \mathbf{Z}_{ij} \geq 0, \end{aligned} \quad (4)$$

Although (4) is sufficient to capture a high-quality anchor similarity matrix, the fuzzy clustering assignment constraint on $\mathbf{Z}^\top \mathbf{1} = \mathbf{1}$ still requires a post-processing step on $\mathbf{Z}$ to obtain a discrete consensus clustering result. To realize a one-step procedure that directly yields the final consensus clustering result, we further introduce a centroid matrix $\mathbf{C} \in \mathbb{R}^{k \times c}$ and a cluster assignment matrix $\mathbf{F} \in \mathbb{R}^{c \times n}$, leading to the following final formulation:

$$\begin{aligned} \min_{\mathbf{P}, \mathbf{A}, \mathbf{Z}, \mathbf{C}, \mathbf{F}} \quad & \|\mathbf{P}^\top \mathbf{H} - \mathbf{AZ}\|_F^2 - \|\mathbf{Z} \odot \mathbf{Z}_c\|_1 + \|\mathbf{Z} - \mathbf{CF}\|_F^2 \\ \text{s.t.} \quad & \mathbf{PP}^\top = \mathbf{I}, \mathbf{A}^\top \mathbf{A} = \mathbf{I}, \mathbf{C}^\top \mathbf{C} = \mathbf{I}, \mathbf{Z}^\top \mathbf{1} = \mathbf{1}, \mathbf{Z}_{ij} \geq 0. \\ & \mathbf{F}_{ij} \in \{0, 1\}, \sum_{i=1}^c \mathbf{F}_{ij} = 1, \forall j = 1, 2, \dots, n, \end{aligned} \quad (5)$$

where $\mathbf{F}_{ij} = 1$ if the j -th sample belongs to the i -th cluster and $\mathbf{F}_{ij} = 0$ otherwise. In this way, the resulting factor matrices admit a strict clustering interpretation, and the discrete consensus clustering result can be directly obtained from $\mathbf{F}$, without any additional post-processing. Compared with YACHT, (5) directly couples the anchor similarity matrix $\mathbf{Z}$ with the discrete assignment matrix $\mathbf{F}$ through $\|\mathbf{Z} - \mathbf{CF}\|_F^2$, and aligns local consensus information with $\mathbf{Z}$ rather than with an intermediate label partition. Therefore, the assignment matrix is optimized within the unified objective instead of being obtained by post-processing.

D. Optimization

To solve (5), we employ the alternating optimization strategy [53], i.e., optimizing the target variable alternately while fixing other variables.

Update P To update the variable $\mathbf{P}$, other variables need to be fixed. The subproblem w.r.t. $\mathbf{P}$ is as follows:

$$\min_{\mathbf{P}} \|\mathbf{P}^\top \mathbf{H} - \mathbf{AZ}\|_F^2 \quad \text{s.t.} \quad \mathbf{PP}^\top = \mathbf{I}.$$

Expanding the Frobenius norm and ignoring constant terms, we rewrite the problem as:

$$\max_{\mathbf{P}} \text{tr}(\mathbf{P}^\top \mathbf{Q}) \quad \text{s.t.} \quad \mathbf{PP}^\top = \mathbf{I}, \quad (6)$$

where $\mathbf{Q} = \mathbf{HZ}^\top \mathbf{A}^\top$. Supposing the singular value decomposition (SVD) of $\mathbf{Q}$ is $\mathbf{U}_Q \Sigma_Q \mathbf{V}_Q^\top$, we can obtain the optimal solution for $\mathbf{P}$ as $\mathbf{U}_Q \mathbf{V}_Q^\top$.

Update A With $\mathbf{P}$, $\mathbf{Z}$, $\mathbf{C}$ and $\mathbf{F}$ being fixed, the subproblem for $\mathbf{A}$ is as follows:

$$\min_{\mathbf{A}} \|\mathbf{P}^\top \mathbf{H} - \mathbf{AZ}\|_F^2 \quad \text{s.t.} \quad \mathbf{A}^\top \mathbf{A} = \mathbf{I}.$$

Similar to the subproblem for $\mathbf{P}$, we transform the subproblem for $\mathbf{A}$ into the following form by expanding the Frobenius norm:

$$\max_{\mathbf{A}} \text{tr}(\mathbf{A}^\top \mathbf{B}) \quad \text{s.t.} \quad \mathbf{A}^\top \mathbf{A} = \mathbf{I}, \quad (7)$$

where $\mathbf{B} = \mathbf{P}^\top \mathbf{H} \mathbf{Z}^\top$. Similarly, we perform the SVD decomposition $\mathbf{B} = \mathbf{U}_B \Sigma_B \mathbf{V}_B^\top$, and then obtain the optimal solution for the variable $\mathbf{A}$ as $\mathbf{U}_B \mathbf{V}_B^\top$.

Update Z With $\mathbf{P}$, $\mathbf{A}$, $\mathbf{C}$ and $\mathbf{F}$ being fixed, the subproblem for $\mathbf{Z}$ is as follows:

$$\begin{aligned} \min_{\mathbf{Z}} \quad & \|\mathbf{P}^\top \mathbf{H} - \mathbf{AZ}\|_F^2 + \|\mathbf{Z} - \mathbf{CF}\|_F^2 - \|\mathbf{Z} \odot \mathbf{Z}_c\|_1 \\ \text{s.t.} \quad & \mathbf{Z}^\top \mathbf{1} = \mathbf{1}, \mathbf{Z}_{ij} \geq 0. \end{aligned} \quad (8)$$

Eq. (8) can be solved using the augmented Lagrange multiplier (ALM) method [54]. The augmented Lagrangian function is:

$$\begin{aligned} \mathcal{L}(\mathbf{Z}) = \quad & \|\mathbf{P}^\top \mathbf{H} - \mathbf{AZ}\|_F^2 + \|\mathbf{Z} - \mathbf{CF}\|_F^2 - \text{tr}(\mathbf{Z}^\top \mathbf{Z}_c) \\ & + \mathbf{J}^\top (\mathbf{Z}^\top \mathbf{1} - \mathbf{1}) + \frac{\mu}{2} \|\mathbf{Z}^\top \mathbf{1} - \mathbf{1}\|_F^2, \end{aligned} \quad (9)$$

where $\mathbf{J} \in \mathbb{R}^{n \times 1}$ is the Lagrange multiplier and μ is the penalty parameter. Next, we take the derivative of (9) and set it to zero. The variable $\mathbf{Z}$ can be updated by:

$$\mathbf{Z} = \mathbf{M}^{-1}(\mu \mathbf{1} \mathbf{1}^\top + 2\mathbf{A}^\top \mathbf{P}^\top \mathbf{H} - \mathbf{1} \mathbf{J}^\top + \mathbf{Z}_c + 2\mathbf{CF}),$$

where $\mathbf{M} = 2\mathbf{A}^\top \mathbf{A} + 2\mathbf{I} + \mu \mathbf{1} \mathbf{1}^\top$. We ensure that $\mathbf{Z}$ satisfies the nonnegativity constraint via $\mathbf{Z} = \max(\mathbf{Z}, 0)$.

Update C With $\mathbf{P}$, $\mathbf{A}$, $\mathbf{Z}$ and $\mathbf{F}$ being fixed, the subproblem for $\mathbf{C}$ is as follows:

$$\min_{\mathbf{C}} \|\mathbf{Z} - \mathbf{CF}\|_F^2 \quad \text{s.t.} \quad \mathbf{C}^\top \mathbf{C} = \mathbf{I}. \quad (10)$$

Eq. (10) can be further rewritten as:

$$\max_{\mathbf{C}} \text{tr}(\mathbf{C}^\top \mathbf{Y}) \quad \text{s.t.} \quad \mathbf{C}^\top \mathbf{C} = \mathbf{I}, \quad (11)$$

where $\mathbf{Y} = \mathbf{ZF}^\top$. We perform the SVD $\mathbf{Y} = \mathbf{U}_Y \Sigma_Y \mathbf{V}_Y^\top$, from which the optimal solution for $\mathbf{C}$ is $\mathbf{U}_Y \mathbf{V}_Y^\top$.

Update F With $\mathbf{P}$, $\mathbf{A}$, $\mathbf{Z}$ and $\mathbf{C}$ being fixed, the subproblem for $\mathbf{F}$ is as follows:

$$\begin{aligned} \min_{\mathbf{F}} \quad & \|\mathbf{Z} - \mathbf{CF}\|_F^2 \\ \text{s.t.} \quad & \mathbf{F}_{ij} \in \{0, 1\}, \sum_{i=1}^c \mathbf{F}_{ij} = 1, \forall j = 1, 2, \dots, n, \end{aligned} \quad (12)$$

Eq. (12) can be further rewritten as:

$$\min_{\mathbf{F}_{:,j}} \|\mathbf{Z}_{:,j} - \mathbf{CF}_{:,j}\|_F^2 \quad \text{s.t.} \quad \mathbf{F}_{:,j} \in \{0, 1\}^c, \|\mathbf{F}_{:,j}\|_1 = 1, \quad (13)$$

The optimal row can be obtained in the following form:

$$i^* = \arg \min_i \|\mathbf{Z}_{:,j} - \mathbf{C}_{:,i}\|_2^2. \quad (14)$$

Update J and μ The Lagrange multipliers and penalty parameters in the ALM method are updated as follows:

$$\mathbf{J} = \mathbf{J} + \mu (\mathbf{Z}^\top \mathbf{1} - \mathbf{1}), \quad \mu = \min(\rho \mu, \mu_{max}), \quad (15)$$

Algorithm 1: DREAM.

Input: m base clustering results, number of clusters c , hyperparameters k' and α .
Output: The final cluster assignment $\mathbf{F}$.

- 1: Initialize $\mathbf{A} = \mathbf{0}$, $\mathbf{Z} = \mathbf{0}$, $\mathbf{C} = \mathbf{I}$, $\mathbf{F} = \mathbf{0}$, $\mathbf{J} = \mathbf{I}$, $\rho = 1.2$, and $\mu = 0.5$. $\mathbf{P}$ is initialized as an orthogonal matrix satisfying $\mathbf{P}\mathbf{P}^\top = \mathbf{I}$. Construct the initial hypergraph $\tilde{\mathbf{H}}$ and obtain the enhanced hypergraph $\mathbf{H}$;
- 2: **while** not converged **do**
- 3: Update $\mathbf{P}$ by solving (6);
- 4: Update $\mathbf{A}$ by solving (7);
- 5: Update $\mathbf{Z}$ by solving (8);
- 6: Update $\mathbf{C}$ by solving (11);
- 7: Update $\mathbf{F}$ by solving (13);
- 8: Update $\mathbf{J}$ and μ by (15);
- 9: Check stopping criteria;
- 10: $\max(\sigma(\mathbf{P}), \sigma(\mathbf{A}), \sigma(\mathbf{F}), \sigma(\mathbf{Z})) \leq 10^{-2}$ with $\sigma(\mathbf{P}) = (||\mathbf{P}_{t+1} - \mathbf{P}_t||_F^2) / ||\mathbf{P}_t||_F^2$;
- 11: **end while**

where $\rho > 1$ is a user-defined constant, and μ_{max} represents the maximum value of μ . The overall process of the algorithm is summarized in Algorithm 1.

E. Complexity Analysis

Time Complexity: In the hypergraph enhancement stage, the hyperedge-level relations are measured by pairwise Jaccard distances. For two hyperedges, the intersection term in the Jaccard distance can be computed from the inner product of their incidence vectors. Therefore, computing all pairwise hyperedge relations costs $\mathcal{O}(s^2n)$. The subsequent high-order random-walk propagation over the hyperedge relation matrix costs $\mathcal{O}(ds^3)$, where d denotes the random-walk order. Therefore, the hypergraph enhancement stage costs $\mathcal{O}(s^2n + ds^3)$.

For variable $\mathbf{P}$, its update process involves SVD decomposition and matrix multiplication, resulting in a time complexity of $\mathcal{O}(s(s')^2 + sk(n + s'))$. For variable $\mathbf{A}$, its update costs $\mathcal{O}(k^2 s' + s'n(s + k))$. For variable $\mathbf{Z}$, the update involves matrix inversion and several matrix multiplications, leading to $\mathcal{O}(k^3 + k^2(s' + n) + ks(s' + n) + kcn)$. For variable $\mathbf{C}$, the update requires computing $\mathbf{Z}\mathbf{F}^\top$ and performing SVD, with a complexity of $\mathcal{O}(kcn + kc^2)$. For variable $\mathbf{F}$, assigning each sample to its nearest centroid costs $\mathcal{O}(kcn)$.

For the optimization stage, the dominant per-iteration cost can be summarized as $\mathcal{O}(n(s'(s + k) + sk + k^2 + kc) + s(s')^2 + k^3 + kc^2)$. This analysis shows that DREAM avoids the $\mathcal{O}(n^2)$ cost of constructing a sample-level co-association matrix. When s, s', k, c , and d are much smaller than n , the sample-dependent terms scale linearly with respect to n . However, the computational cost is not independent of s . When the ensemble contains many base clusters, the s -dependent terms, especially $\mathcal{O}(s^2n)$ and $\mathcal{O}(ds^3)$, may become the computational bottleneck.

Space Complexity: For DREAM, the main storage overhead comes from the enhanced hypergraph, the hyperedge-level relation matrix, and the optimization variables. The enhanced

TABLE I
MAIN NOTATIONS AND DESCRIPTIONS

Notations	Descriptions
n, c	Number of samples and clusters.
s, s'	Number of base clusters and representative base clusters.
k	Number of anchors.
$\mathbf{H} \in \mathbb{R}^{s \times n}$	The enhanced hypergraph incidence matrix.
$\tilde{\mathbf{O}} \in \mathbb{R}^{s \times s}$	The high-order similarity matrix of hyperedges.
$\mathbf{P} \in \mathbb{R}^{s \times s'}$	The projection matrix.
$\mathbf{A} \in \mathbb{R}^{s' \times k}$	The anchor matrix.
$\mathbf{Z} \in \mathbb{R}^{k \times n}$	The anchor similarity matrix.
$\mathbf{C} \in \mathbb{R}^{k \times c}$	The centroid matrix.
$\mathbf{F} \in \{0, 1\}^{c \times n}$	The cluster assignment matrix.
$\mathbf{Z}_c \in \{0, 1\}^{k \times n}$	The local consensus information indicator matrix.

TABLE II
CHARACTERISTICS OF ALL DATASETS

No.	Dataset	Instance	Feature	Class
D1	Blobs-Rings	1000000	2	6
D2	Website Phishing	1353	9	3
D3	Obesity	2111	16	7
D4	ALOI100	10000	128	100
D5	Shuttle	58000	9	7
D6	KannadaMNIST	70000	784	10
D7	SensIT-acoustic	98528	50	3
D8	EMNIST-train	240000	784	10
D9	Winnipeg	325834	174	7
D10	Covtype	581012	54	7

hypergraph incidence matrix $\mathbf{H}$ requires $\mathcal{O}(sn)$ space, while the hyperedge-level relation matrix requires $\mathcal{O}(s^2)$ space. The variables $\mathbf{P}$, $\mathbf{A}$, $\mathbf{Z}$, $\mathbf{Z}_c$, $\mathbf{C}$, and $\mathbf{F}$ further require $\mathcal{O}(ss' + s'k + kn + kc + cn)$ space. Therefore, the overall space complexity is $\mathcal{O}(sn + s^2)$. This shows that DREAM avoids the $\mathcal{O}(n^2)$ memory cost of CA matrix-based methods, while its memory usage still depends on the number of hyperedges through the $\mathcal{O}(sn)$ and $\mathcal{O}(s^2)$ terms.

IV. EXPERIMENT

All experiments were conducted on a PC with an AMD 7763 64-Core Processor 2.45 GHz CPU, 128 GB memory, running Windows 11 Professional. The algorithms were implemented in MATLAB (R2024b). The code is available at <https://github.com/middle258/DREAM>.

A. Experimental Setup

We conducted extensive experiments on 1 synthetic dataset (**Blobs-Rings**) and 9 real datasets. Characteristics of these datasets are provided in Table II. Some methods [24] provide relevant base clustering results, which we directly adopt for convenience. For the other datasets^{1,2,3} [55], we randomly run the K-means method 100 times to generate the base clustering result set [56]. Following many ensemble methods, we set the

¹<http://archive.ics.uci.edu/datasets>

²<https://github.com/k-atif/SBSC>

³<https://www.nist.gov/itl/products-and-services/emnist-dataset>

TABLE III

CLUSTERING PERFORMANCE IS MEASURED BY ACC (%). THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD. ‘-’ INDICATES EXCESSIVE RUNTIME. ‘OM’ MEANS OUT OF MEMORY

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
Base clusterings (average)	81.59 ± 12.64	22.82 ± 12.16	28.12 ± 7.42	64.80 ± 1.76	47.89 ± 4.52	63.33 ± 3.60	46.59 ± 1.69	55.43 ± 4.09	34.93 ± 1.35	29.96 ± 3.08
SEC [KDD, 2015]	72.71 ± 11.35	48.15 ± 6.18	34.74 ± 3.08	54.49 ± 2.82	45.54 ± 9.73	58.25 ± 6.62	47.11 ± 2.85	50.28 ± 6.57	36.22 ± 3.52	32.85 ± 4.10
PTA-AL [TKDE, 2016]	71.41 ± 12.35	49.13 ± 4.30	41.74 ± 1.07	57.40 ± 1.14	50.72 ± 6.14	60.26 ± 5.28	47.00 ± 2.00	54.33 ± 3.91	36.02 ± 3.05	33.58 ± 4.77
PTA-CL [TKDE, 2016]	86.63 ± 9.36	50.49 ± 4.09	40.95 ± 1.25	57.81 ± 1.20	49.18 ± 6.69	62.64 ± 1.65	49.13 ± 4.13	55.39 ± 3.02	35.38 ± 1.41	33.01 ± 4.30
PTGP [TKDE, 2016]	83.97 ± 8.04	52.43 ± 3.42	42.36 ± 1.86	57.41 ± 1.29	45.13 ± 9.81	63.12 ± 1.59	47.29 ± 1.54	54.99 ± 3.13	35.17 ± 1.45	32.62 ± 4.74
LWGP [TCYB, 2018]	81.17 ± 8.33	58.19 ± 1.73	41.77 ± 1.01	68.65 ± 0.98	49.97 ± 0.82	61.13 ± 2.78	48.07 ± 0.12	55.05 ± 4.37	34.41 ± 0.61	29.38 ± 0.77
DREC [Neuro., 2019]	87.27 ± 6.51	52.76 ± 2.98	41.53 ± 0.72	65.53 ± 1.28	48.66 ± 2.35	62.55 ± 1.59	48.49 ± 2.73	53.66 ± 3.09	35.05 ± 2.00	30.43 ± 1.91
USENC [TKDE, 2020]	85.13 ± 8.87	56.37 ± 3.87	41.04 ± 1.07	69.95 ± 1.92	48.18 ± 3.59	62.92 ± 3.26	47.85 ± 0.15	54.68 ± 5.08	33.56 ± 1.05	29.18 ± 0.18
TRCE [AAAI, 2021]	OM	58.76 ± 0.39	41.43 ± 0.75	70.21 ± 0.90	-	-	-	OM	OM	OM
ECPCS-MC [TSMC, 2021]	87.48 ± 8.96	57.04 ± 2.58	41.76 ± 0.74	67.53 ± 1.02	54.47 ± 12.53	62.06 ± 0.11	47.58 ± 1.24	55.95 ± 3.60	34.06 ± 0.00	29.38 ± 0.77
CESHL [Inf. Fusion, 2022]	OM	58.68 ± 0.34	41.89 ± 0.64	-	-	-	-	OM	OM	OM
ECCMS [TNNLS, 2024]	OM	55.97 ± 4.67	41.72 ± 1.30	65.40 ± 1.60	-	-	-	OM	OM	OM
CEAM [TKDE, 2024]	OM	52.76 ± 6.32	40.54 ± 1.91	67.10 ± 1.10	-	-	-	OM	OM	OM
GPEC [ICML, 2025]	OM	57.07 ± 4.86	42.28 ± 2.15	66.71 ± 1.92	-	-	-	OM	OM	OM
CEHM [AAAI, 2025]	84.37 ± 8.96	58.50 ± 0.75	40.95 ± 0.54	68.02 ± 0.73	48.72 ± 2.38	61.67 ± 1.58	47.23 ± 1.66	55.64 ± 1.55	34.07 ± 0.00	32.73 ± 2.34
FSEC [Inf. Fusion, 2025]	85.93 ± 8.43	57.07 ± 3.45	41.41 ± 0.75	70.80 ± 1.11	48.83 ± 2.41	63.07 ± 2.68	48.07 ± 0.12	56.11 ± 3.78	34.03 ± 0.00	29.13 ± 0.04
YACHT [IJCAI, 2025]	85.63 ± 9.75	59.08 ± 5.54	42.59 ± 1.41	71.94 ± 1.18	47.99 ± 2.10	66.95 ± 3.38	51.10 ± 2.05	58.35 ± 3.89	35.17 ± 1.76	30.38 ± 1.07
UDBGL [TNNLS, 2024]	-	49.66 ± 5.57	29.08 ± 12.49	61.98 ± 1.64	44.30 ± 8.55	35.14 ± 12.83	46.62 ± 2.34	-	-	-
DREAM (Ours)	88.00 ± 8.99	64.15 ± 4.28	42.26 ± 0.67	73.78 ± 1.19	61.89 ± 6.59	67.89 ± 3.09	51.55 ± 1.81	59.14 ± 1.03	36.30 ± 3.13	34.79 ± 3.02

TABLE IV

CLUSTERING PERFORMANCE IS MEASURED BY ARI (%). THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD. ‘-’ INDICATES EXCESSIVE RUNTIME. ‘OM’ MEANS OUT OF MEMORY

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
Base clusterings (average)	68.15 ± 20.95	8.59 ± 4.01	18.49 ± 4.31	56.66 ± 1.74	24.37 ± 3.71	48.42 ± 2.05	5.85 ± 2.47	37.88 ± 2.66	13.16 ± 1.90	4.13 ± 2.03
SEC [KDD, 2015]	62.70 ± 14.81	2.69 ± 3.87	14.19 ± 4.18	25.61 ± 7.68	7.34 ± 8.96	40.92 ± 7.97	4.57 ± 3.71	32.47 ± 4.90	11.78 ± 4.02	0.89 ± 5.08
PTA-AL [TKDE, 2016]	64.54 ± 13.66	6.93 ± 3.98	21.70 ± 0.79	51.56 ± 0.73	23.46 ± 7.22	47.44 ± 3.45	2.84 ± 2.84	37.18 ± 3.44	14.64 ± 2.68	6.63 ± 2.21
PTA-CL [TKDE, 2016]	76.82 ± 13.18	8.53 ± 3.60	21.61 ± 0.93	51.93 ± 0.79	18.09 ± 9.49	48.52 ± 1.58	5.00 ± 4.73	38.71 ± 2.26	13.95 ± 1.95	6.78 ± 2.69
PTGP [TKDE, 2016]	71.81 ± 12.96	9.91 ± 2.99	21.93 ± 1.41	52.23 ± 0.96	20.37 ± 6.06	49.08 ± 1.25	7.09 ± 1.60	38.00 ± 2.83	13.92 ± 1.71	6.74 ± 2.69
LWGP [TCYB, 2018]	66.44 ± 12.26	17.54 ± 2.93	21.29 ± 0.27	59.73 ± 0.70	27.05 ± 0.00	47.74 ± 2.91	7.91 ± 0.13	38.11 ± 4.07	12.46 ± 1.02	3.57 ± 0.19
DREC [Neuro., 2019]	77.68 ± 9.82	7.32 ± 2.76	21.27 ± 0.39	55.74 ± 2.57	25.27 ± 2.96	47.69 ± 2.23	7.40 ± 2.89	35.65 ± 2.64	13.28 ± 1.70	4.00 ± 0.74
USENC [TKDE, 2020]	73.21 ± 10.54	16.43 ± 5.69	20.96 ± 0.69	61.09 ± 1.68	23.51 ± 3.35	48.63 ± 2.78	7.55 ± 0.24	38.18 ± 4.26	12.06 ± 0.31	3.42 ± 0.30
TRCE [AAAI, 2021]	OM	17.12 ± 4.07	21.03 ± 0.46	61.36 ± 0.69	-	-	-	OM	OM	OM
ECPCS-MC [TSMC, 2021]	76.87 ± 14.47	16.46 ± 7.66	21.45 ± 0.36	59.17 ± 1.09	32.25 ± 13.31	48.41 ± 0.19	7.39 ± 1.30	39.12 ± 2.92	12.03 ± 0.00	3.63 ± 0.25
CESHL [Inf. Fusion, 2022]	OM	18.26 ± 4.50	20.97 ± 0.78	-	-	-	-	OM	OM	OM
ECCMS [TNNLS, 2024]	OM	13.46 ± 5.40	21.24 ± 0.50	57.30 ± 0.40	-	-	-	OM	OM	OM
CEAM [TKDE, 2024]	OM	12.92 ± 6.34	20.67 ± 1.11	59.10 ± 0.90	-	-	-	OM	OM	OM
GPEC [ICML, 2025]	OM	13.66 ± 4.66	23.54 ± 2.95	57.11 ± 1.04	-	-	-	OM	OM	OM
CEHM [AAAI, 2025]	72.28 ± 14.27	17.66 ± 3.51	20.35 ± 0.54	59.08 ± 0.65	26.24 ± 2.55	47.95 ± 1.07	7.02 ± 1.75	38.67 ± 1.47	12.03 ± 0.00	3.64 ± 0.40
FSEC [Inf. Fusion, 2025]	73.33 ± 13.51	17.70 ± 5.23	21.24 ± 0.33	61.85 ± 0.84	24.84 ± 2.95	48.97 ± 1.90	7.91 ± 0.13	39.36 ± 3.06	12.01 ± 0.00	3.60 ± 0.32
YACHT [IJCAI, 2025]	73.52 ± 15.78	17.05 ± 8.61	23.93 ± 1.33	62.55 ± 1.17	20.13 ± 1.61	50.47 ± 2.56	7.42 ± 1.69	40.94 ± 3.33	15.67 ± 1.23	5.48 ± 0.64
UDBGL [TNNLS, 2024]	-	6.41 ± 4.11	11.88 ± 12.67	41.96 ± 10.03	14.71 ± 5.45	16.13 ± 13.05	3.96 ± 1.28	-	-	-
DREAM (Ours)	77.74 ± 14.57	19.94 ± 10.87	24.20 ± 1.49	63.63 ± 1.21	33.90 ± 9.49	50.51 ± 2.44	8.99 ± 1.45	41.50 ± 1.55	16.41 ± 2.32	4.77 ± 0.79

ensemble size to 20 and conduct 10 repeated experiments with different base clustering combinations, reporting the mean and standard deviation. All compared methods use the same base clustering combinations. In DREAM, we follow the settings of the original paper [57], with the maximum order of hyperedge random walk set to $d = 10$. The hyperparameters are set as follows: the number of anchors $k = k' + c$, where $k' \in \{-1, 0, 1, 2, 3\}$ and c is the number of clusters; the similarity threshold $\alpha \in [0.9 : 0.01 : 1]$. We perform grid search over two hyperparameters to obtain the best results. To ensure a fair comparison, all compared methods are exhaustively searched over all parameter combinations given in their respective papers, and their optimal results are reported, ensuring that each algorithm is evaluated under its best settings. We compare DREAM with 17 representative clustering methods, covering state-of-the-art ensemble clustering methods, fast ensemble clustering methods, and one-step multi-view clustering methods. • SOTA ensemble methods: ECCMS [50], CEAM [5], CESHL [32], TRCE [58], DREC [59], ECPCS-MC [56], GPEC [24], YACHT [13]. •

Fast ensemble methods: LWGP [57], USENC [11], SEC [60], PTA-AL/CL [52], PTGP [52], CEHM [5], FSEC [8]. • One-step multi-view clustering method: UDBGL [61].

All methods are evaluated using four popular clustering metrics: Accuracy (ACC), Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and Purity. For all evaluation metrics, larger values indicate better algorithm performance.

B. Experimental Results

Tables III and IV list the comparative experimental results of DREAM in terms of ACC and ARI. Tables V and VI list the experimental results in terms of NMI and Purity. The best experimental results are highlighted in bold. ‘-’ indicates excessive runtime. ‘OM’ means out of memory.

The experimental results yield several key observations: (i) The ensemble methods consistently surpass the average performance of the individual base clusterings (generated by K-means) across most datasets. This demonstrates the overall

TABLE V
CLUSTERING PERFORMANCE IS MEASURED BY NMI (%). THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD. '-' INDICATES EXCESSIVE RUNTIME. 'OM' MEANS OUT OF MEMORY

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
Base clusterings (average)	80.71 ± 12.55	18.65 ± 2.91	30.57 ± 4.11	83.76 ± 0.52	45.42 ± 2.97	59.90 ± 0.76	7.65 ± 2.08	49.28 ± 1.62	22.80 ± 2.64	13.99 ± 2.03
SEC [KDD, 2015]	78.94 ± 8.95	5.45 ± 3.54	21.94 ± 4.45	78.14 ± 1.87	24.08 ± 7.99	55.54 ± 3.15	6.39 ± 4.06	46.26 ± 2.44	18.56 ± 4.29	10.03 ± 3.73
PTA-AL [TKDE, 2016]	79.40 ± 9.01	6.31 ± 2.42	28.43 ± 1.09	82.01 ± 0.23	40.82 ± 7.02	59.57 ± 2.17	6.02 ± 4.25	49.42 ± 1.96	23.32 ± 2.59	12.42 ± 2.64
PTA-CL [TKDE, 2016]	86.89 ± 7.27	6.90 ± 1.90	28.11 ± 1.32	81.95 ± 0.23	35.16 ± 6.29	60.14 ± 0.81	7.45 ± 3.90	50.19 ± 1.34	23.88 ± 2.77	11.91 ± 2.30
PTGP [TKDE, 2016]	83.84 ± 7.41	7.62 ± 1.48	28.79 ± 1.69	82.10 ± 0.33	41.44 ± 5.44	60.59 ± 0.56	8.16 ± 0.46	49.59 ± 1.61	23.31 ± 2.34	12.35 ± 2.77
LWGP [TCYB, 2018]	81.05 ± 6.79	14.05 ± 2.84	28.09 ± 0.48	84.84 ± 0.21	47.62 ± 0.31	60.37 ± 0.95	7.97 ± 0.04	49.31 ± 2.19	22.11 ± 2.05	14.43 ± 0.38
DREC [Neuro., 2019]	86.73 ± 5.61	8.08 ± 4.97	27.97 ± 0.59	84.08 ± 0.41	45.74 ± 2.94	59.37 ± 1.72	7.32 ± 2.60	46.27 ± 1.87	21.24 ± 1.82	14.00 ± 0.98
USENC [TKDE, 2020]	84.59 ± 7.49	13.30 ± 4.17	27.61 ± 0.79	84.98 ± 0.42	45.35 ± 1.71	59.84 ± 2.21	7.67 ± 0.02	49.51 ± 3.31	20.91 ± 0.55	13.88 ± 1.23
TRCE [AAAI, 2021]	OM	14.21 ± 2.92	27.94 ± 0.60	85.30 ± 0.21	-	-	-	OM	OM	OM
ECPCS-MC [TSMC, 2021]	86.37 ± 8.17	13.29 ± 6.32	28.18 ± 0.42	84.64 ± 0.31	48.41 ± 5.38	60.37 ± 0.15	8.04 ± 0.32	50.27 ± 1.60	21.16 ± 0.00	14.46 ± 0.38
CESHL [Inf. Fusion, 2022]	OM	14.53 ± 3.37	28.26 ± 0.72	-	-	-	-	OM	OM	OM
ECCMS [TNNLS, 2024]	OM	10.91 ± 3.54	28.27 ± 0.38	84.50 ± 0.30	-	-	-	OM	OM	OM
CEAM [TKDE, 2024]	OM	10.63 ± 4.63	27.25 ± 0.96	83.80 ± 0.20	-	-	-	OM	OM	OM
GPEC [ICML, 2025]	OM	12.99 ± 4.12	30.42 ± 1.93	83.66 ± 0.33	-	-	-	OM	OM	OM
CEHM [AAAI, 2025]	83.81 ± 8.01	14.65 ± 2.56	27.37 ± 0.70	84.77 ± 0.25	46.99 ± 1.32	59.80 ± 0.80	8.15 ± 0.42	49.60 ± 1.35	21.15 ± 0.00	14.52 ± 0.23
FSEC [Inf. Fusion, 2025]	84.63 ± 7.56	14.24 ± 3.84	27.98 ± 0.39	85.18 ± 0.19	46.05 ± 1.60	60.43 ± 1.07	7.97 ± 0.04	50.46 ± 1.84	21.16 ± 0.00	14.33 ± 0.15
YACHT [IJCAI, 2025]	84.37 ± 9.09	14.48 ± 4.96	30.39 ± 1.47	85.06 ± 0.33	43.99 ± 1.72	60.57 ± 1.15	10.24 ± 1.01	50.42 ± 1.36	24.08 ± 1.86	14.09 ± 0.90
UDBGL [TNNLS, 2024]	-	6.92 ± 3.86	15.63 ± 15.25	81.41 ± 2.32	36.60 ± 4.42	35.96 ± 15.11	9.03 ± 2.85	-	-	-
DREAM (Ours)	86.98 ± 8.59	17.81 ± 5.95	30.39 ± 1.31	85.38 ± 0.36	49.00 ± 4.25	60.30 ± 1.40	11.02 ± 1.68	50.95 ± 0.93	23.51 ± 2.68	14.64 ± 0.67

TABLE VI
CLUSTERING PERFORMANCE IS MEASURED BY PURITY (%). THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD. '-' INDICATES EXCESSIVE RUNTIME. 'OM' MEANS OUT OF MEMORY

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
Base clusterings (average)	81.62 ± 12.61	78.11 ± 4.67	51.07 ± 7.45	68.23 ± 1.54	94.21 ± 2.53	67.02 ± 2.12	51.58 ± 1.29	58.01 ± 3.04	47.56 ± 2.68	54.09 ± 1.62
SEC [KDD, 2015]	76.14 ± 11.08	56.95 ± 5.00	37.09 ± 2.86	57.98 ± 2.75	83.74 ± 3.07	60.05 ± 5.66	51.37 ± 1.47	52.36 ± 5.01	44.11 ± 4.46	52.38 ± 1.57
PTA-AL [TKDE, 2016]	74.13 ± 10.36	59.73 ± 2.82	43.41 ± 0.97	59.35 ± 1.16	91.35 ± 3.61	64.59 ± 5.00	50.98 ± 1.28	57.13 ± 3.45	48.05 ± 2.77	52.17 ± 1.49
PTA-CL [TKDE, 2016]	87.13 ± 8.17	60.96 ± 2.34	43.02 ± 0.99	60.10 ± 0.96	89.34 ± 2.66	66.42 ± 1.04	52.52 ± 2.12	57.88 ± 2.87	48.94 ± 2.76	51.91 ± 1.46
PTGP [TKDE, 2016]	83.97 ± 8.04	61.98 ± 1.70	43.78 ± 1.21	58.59 ± 1.20	92.87 ± 3.06	66.54 ± 0.79	52.14 ± 1.14	57.65 ± 2.64	48.59 ± 2.43	52.25 ± 1.61
LWGP [TCYB, 2018]	81.57 ± 7.89	67.35 ± 1.95	43.29 ± 0.54	69.97 ± 1.05	94.73 ± 0.00	65.24 ± 2.80	52.74 ± 0.12	57.32 ± 3.65	46.53 ± 1.33	53.93 ± 0.79
DREC [Neuro., 2019]	87.27 ± 6.51	61.35 ± 2.97	43.46 ± 0.82	67.98 ± 1.26	94.40 ± 1.17	65.62 ± 2.65	52.72 ± 1.37	55.55 ± 2.17	46.48 ± 1.85	54.99 ± 1.89
USENC [TKDE, 2020]	82.73 ± 9.58	66.44 ± 4.64	42.68 ± 0.89	72.40 ± 1.65	94.74 ± 0.13	65.94 ± 2.46	52.34 ± 0.17	56.80 ± 5.25	45.94 ± 0.17	53.70 ± 1.00
TRCE [AAAI, 2021]	OM	66.93 ± 3.09	43.06 ± 0.89	72.95 ± 0.77	-	-	-	OM	OM	OM
ECPCS-MC [TSMC, 2021]	87.48 ± 8.96	66.66 ± 5.47	43.38 ± 0.59	69.95 ± 1.15	94.77 ± 0.00	66.17 ± 0.10	52.37 ± 0.85	58.49 ± 3.02	45.98 ± 0.13	54.04 ± 0.77
CESHL [Inf. Fusion, 2022]	OM	68.08 ± 3.73	43.17 ± 0.85	-	-	-	-	OM	OM	OM
ECCMS [TNNLS, 2024]	OM	64.68 ± 3.64	43.24 ± 0.86	66.60 ± 1.80	-	-	-	OM	OM	OM
CEAM [TKDE, 2024]	OM	64.18 ± 4.39	42.68 ± 1.23	69.70 ± 0.90	-	-	-	OM	OM	OM
GPEC [ICML, 2025]	OM	67.37 ± 4.62	43.92 ± 1.50	68.55 ± 1.64	-	-	-	OM	OM	OM
CEHM [AAAI, 2025]	84.37 ± 8.96	68.01 ± 1.76	42.46 ± 0.85	69.26 ± 0.92	94.77 ± 0.00	65.37 ± 0.87	52.14 ± 1.14	56.63 ± 2.47	45.95 ± 0.00	52.76 ± 0.87
FSEC [Inf. Fusion, 2025]	85.93 ± 8.43	67.49 ± 4.31	43.11 ± 0.46	73.04 ± 0.86	94.77 ± 0.11	66.89 ± 1.38	52.74 ± 0.12	58.60 ± 3.09	45.98 ± 0.00	53.90 ± 0.52
YACHT [IJCAI, 2025]	85.63 ± 9.75	68.34 ± 4.08	44.07 ± 1.27	74.07 ± 0.95	94.09 ± 1.34	68.60 ± 2.22	53.14 ± 1.07	60.03 ± 3.71	48.72 ± 1.10	54.62 ± 2.57
UDBGL [TNNLS, 2024]	-	60.18 ± 4.61	30.16 ± 13.62	64.78 ± 1.75	88.93 ± 3.38	35.45 ± 12.95	49.97 ± 0.00	-	-	-
DREAM (Ours)	88.00 ± 8.99	70.64 ± 4.79	43.51 ± 0.49	75.37 ± 1.26	94.83 ± 1.03	68.79 ± 2.26	52.71 ± 0.00	60.65 ± 1.33	49.61 ± 2.44	55.38 ± 2.52

effectiveness of the ensemble strategy in yielding a more robust consensus. Specifically, in terms of ACC, DREAM achieves an average improvement of 10.43% over the average base clustering performance across all datasets. (ii) Whether on the synthetic dataset Blobs-Rings or on real-world datasets, DREAM demonstrates strong performance advantages. Moreover, across various datasets of different scales, DREAM consistently shows significant superiority, highlighting its excellent scalability. In particular, in terms of ACC, DREAM achieves the best performance on most datasets. (iii) For methods based on the CA matrix, such as ECCMS, GPEC, and CEAM, their time complexity prevents them from scaling to large-scale datasets. Even on the smaller datasets, GPEC attains the best performance only once in terms of NMI metric. For methods built on sample-cluster matrices or hypergraphs, such as DREC and CESH, none of them achieves the best result on any dataset. This indicates that the hypergraph enhancement and reconstruction strategy adopted by DREAM is more effective. It is worth noting that DREC preserves part of the local consensus information via microclusters; however,

its exploration of high-order relationships among base clusters remains limited, making it difficult to surpass the performance of DREAM. (iv) The DREAM method surpasses the YACHT method on most metrics across the majority of datasets, demonstrating the effectiveness of the extended version. On a few datasets, however, a performance drop is observed. A possible explanation is that the proposed alignment strategy reinforces the embedding of local consensus information and, unlike YACHT, does not align through the label partition induced by the hypergraph regularizer, thereby preserving the local consensus information as much as possible. Nevertheless, because this local consensus information is ultimately derived from the base clustering results, the correctness of the sample-to-cluster memberships cannot be fully guaranteed. Overall, by operating directly on the anchor similarity matrix, the alignment strategy tends to improve its accuracy and quality in most cases, as evidenced by the results on large-scale datasets such as Blobs-Rings and Covtype.

We further perform the Friedman test on the methods shown in Fig. 3. The blue dots denote the average ranks, where a lower

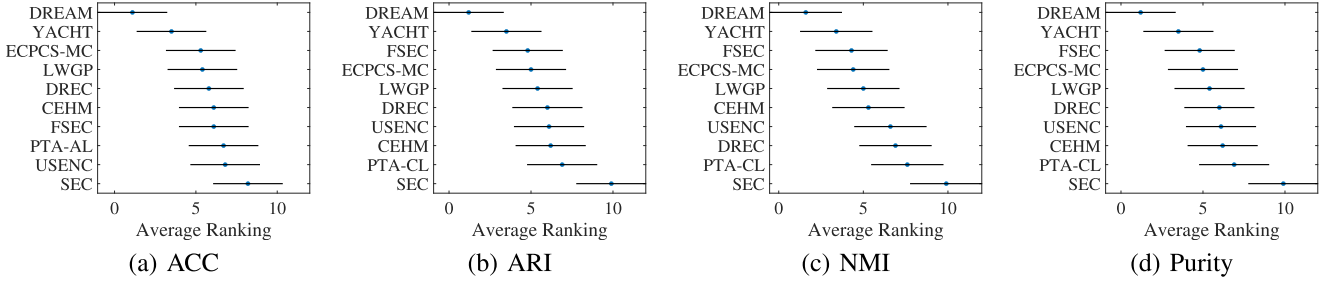


Fig. 3. Average ranking of different clustering methods based on the Friedman test.

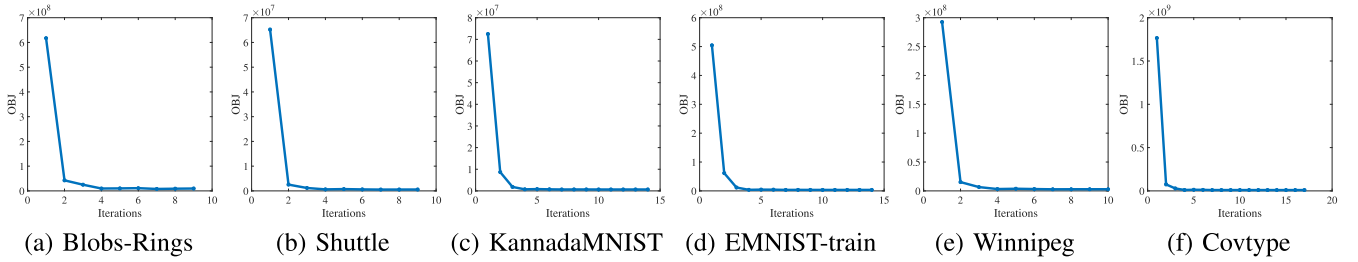


Fig. 4. Convergence curves of the proposed DREAM.

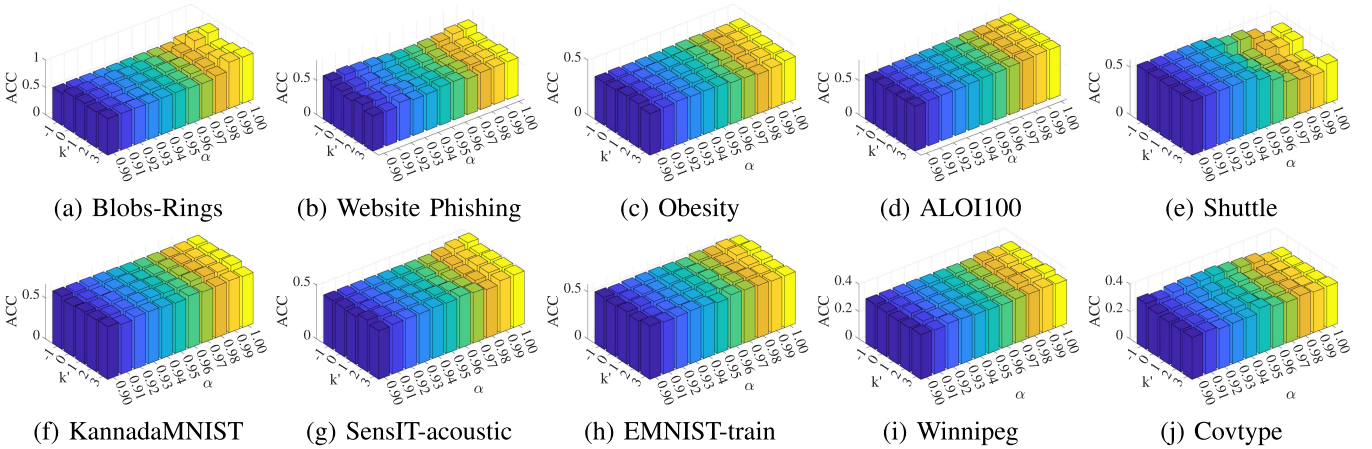


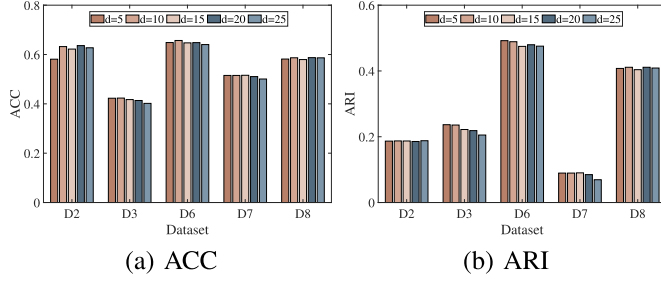
Fig. 5. ACC w.r.t. k' and α .

rank indicates better overall performance. Non-overlapping confidence intervals indicate a statistically significant difference between two methods. As shown in Fig. 3, DREAM obtains the lowest average rank on ACC, ARI, NMI, and Purity. Its ranking also shows clear advantages over several competing methods, further confirming the robustness of DREAM across different datasets. In addition, as shown in Fig. 4, we experimentally verify the convergence of the proposed DREAM, and we observe that DREAM converges after only a small number of iterations.

C. Parameter Study

Fig. 5 shows the impact of different parameter combinations on the accuracy of the DREAM method. DREAM mainly has

two hyperparameters: the number of anchors $k = c + k'$ and the similarity threshold parameter α . The number of anchors k affects hypergraph reconstruction, while the similarity threshold α influences hypergraph enhancement. We observe that, on most datasets, the DREAM method is not highly sensitive to parameter combinations, indicating that DREAM exhibits good robustness. For certain datasets, such as SensIT-acoustic and Blobs-Rings, the optimal results are concentrated in regions with larger values of the similarity threshold α ($\alpha > 0.98$). This suggests that for some datasets, extremely high-similarity base clusters are required for enhancement to achieve high accuracy. In other words, a larger α restricts the enhancement of the hypergraph to clusters with higher similarity, which indicates that these datasets require stronger cluster reliability. The only


 Fig. 6. Sensitivity analysis of the random walk order d .

exception is the Shuttle dataset, for which the method maintains good performance only when the similarity threshold lies in the range 0.90–0.94. This suggests that, for this dataset, the hyperedge structure must undergo substantial adjustments to approach the ground truth. Moreover, we observe that even on the million-scale Blobs-Rings dataset, reconstructing the hypergraph with only a small number of anchors can still yield very high clustering accuracy, which verifies the effectiveness of the reconstruction. Overall, based on the above observations, we recommend setting the number of anchors to c or $c - 1$, i.e., $k' = 0$ or $k' = -1$, and using a relatively large similarity threshold with $\alpha > 0.98$.

Although the maximum random walk order d is fixed to the default value $d = 10$ in our main experiments, we further conduct an additional sensitivity analysis to examine whether this default setting is reliable. Specifically, we vary d within $\{5, 10, 15, 20, 25\}$ and report the ACC and ARI results in Fig. 6. The results show that DREAM is generally stable with respect to different values of d on most datasets, indicating that the hypergraph enhancement stage is not overly dependent on a specific random walk order. Nevertheless, on a few datasets, either an overly small or excessively large value of d may lead to performance degradation. This is reasonable because a small d may be insufficient to capture high-order relations among hyperedges, whereas a large d may cause over-propagation and introduce unreliable relations from noisy base clusters. Overall, these observations support the use of the default setting $d = 10$ in our experiments.

D. Time Comparison

Fig. 7 shows the runtime comparison of different methods. The fastest ensemble methods are USENC and FSEC, as their selected anchors are fixed and remain unchanged once determined. In contrast, DREAM requires dynamically updating anchors during the hypergraph reconstruction process, which inevitably incurs additional time costs. However, dynamic anchor learning produces a higher-quality anchor similarity matrix and improves clustering accuracy. For CA matrix-based methods such as GPEC, it is evident that they struggle to complete the ensemble on large-scale datasets. Their running time is still significantly higher than that of other ensemble clustering methods. For methods that introduce microclusters, such as PTGP and DREC, their runtime depends on the number of microclusters. If the microclusters exhibit a fragmented, point-like distribution, the

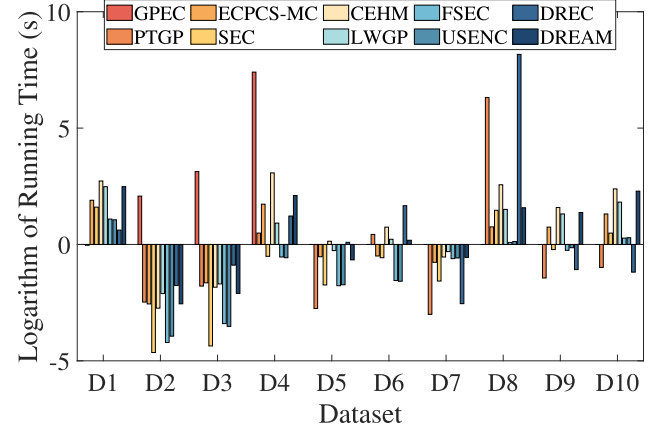


Fig. 7. The running time comparison of different methods.

runtime increases sharply (e.g., on the EMNIST dataset). In contrast, DREAM is not affected by this issue and is overall slightly faster than CEHM. Although DREAM is not the fastest ensemble clustering method, in practice, a modest increase in runtime is acceptable in exchange for higher accuracy.

E. Ablation Study

Table VII reports the ablation results of four degraded variants of DREAM on four evaluation metrics. We mainly examine the effectiveness of the mapping matrix $\mathbf{P}$, the hypergraph enhancement strategy, the local consensus information alignment strategy, and the one-step assignment term. Specifically, “w/o Matrix $\mathbf{P}$ ” denotes the variant without the mapping matrix $\mathbf{P}$, “w/o H-enhanced” denotes the variant that does not apply the hypergraph enhancement strategy, “w/o $\|\mathbf{Z} \odot \mathbf{Z}_c\|_1$ ” denotes the variant without the local consensus information alignment term, and “w/o $\|\mathbf{Z} - \mathbf{CF}\|_F^2$ ” denotes the variant that removes the one-step assignment term and applies k -means to $\mathbf{Z}^\top$ to obtain the final clustering result. We observe that, on most datasets, these degraded variants perform worse than the full DREAM model, which confirms the effectiveness of the key components.

Overall, removing the hypergraph enhancement strategy leads to a pronounced performance degradation, indicating that the quality of the input hypergraph is crucial for DREAM, since it directly influences the hypergraph reconstruction term $\|\mathbf{P}^\top \mathbf{H} - \mathbf{AZ}\|_F^2$ and, consequently, the quality of the anchor similarity matrix. In addition, the variant w/o $\|\mathbf{Z} - \mathbf{CF}\|_F^2$ is generally inferior to the full model. This demonstrates the necessity of the one-step ensemble clustering design. By directly coupling $\mathbf{Z}$ with the discrete assignment matrix $\mathbf{F}$ through $\|\mathbf{Z} - \mathbf{CF}\|_F^2$, the full DREAM encourages the learned anchor similarity matrix to be more consistent with the final clustering structure. Discarding the local consensus alignment term or the mapping matrix $\mathbf{P}$ causes relatively smaller performance drops, and even leads to slight improvements on a few datasets. This may be because the local consensus information on these specific datasets is less reliable or partially redundant with the structural information captured by the enhanced hypergraph reconstruction. In

TABLE VII
ABLATION EXPERIMENTAL RESULTS ON THE FOUR METRICS (%). THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD

Metric	Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
ACC	w/o Matrix P	87.37 ± 9.00	63.78 ± 4.55	41.21 ± 2.11	72.27 ± 1.22	50.68 ± 4.68	67.80 ± 2.89	51.84 ± 3.48	58.51 ± 5.26	36.94 ± 2.80	32.76 ± 3.58
	w/o H-enhanced	87.37 ± 9.03	60.94 ± 4.93	41.04 ± 2.63	73.11 ± 0.72	41.39 ± 1.12	65.48 ± 3.14	51.50 ± 1.57	58.14 ± 0.96	36.03 ± 2.18	32.46 ± 4.91
	w/o $\ Z \odot Z_c\ _1$	87.28 ± 9.16	63.04 ± 6.64	42.43 ± 0.62	72.07 ± 0.80	58.60 ± 5.08	64.69 ± 3.06	49.95 ± 2.44	58.70 ± 1.44	35.88 ± 2.18	33.72 ± 5.66
	w/o $\ Z - CF\ _F^2$	87.38 ± 9.02	55.00 ± 5.67	42.15 ± 2.28	72.65 ± 0.99	46.97 ± 5.11	66.56 ± 3.81	51.97 ± 4.91	58.46 ± 2.07	33.36 ± 0.81	29.49 ± 0.74
	DREAM	88.00 ± 8.99	64.15 ± 4.28	42.26 ± 0.67	73.78 ± 1.19	61.89 ± 6.59	67.89 ± 3.09	51.55 ± 1.81	59.14 ± 1.03	36.30 ± 3.13	34.79 ± 3.02
ARI	w/o Matrix P	76.76 ± 14.40	19.66 ± 8.44	21.96 ± 1.74	61.93 ± 1.52	24.13 ± 2.07	49.59 ± 1.52	8.49 ± 2.90	41.04 ± 0.81	15.31 ± 1.81	5.07 ± 1.82
	w/o H-enhanced	76.77 ± 14.49	15.60 ± 11.00	21.55 ± 2.96	63.13 ± 0.86	18.59 ± 0.37	49.27 ± 2.39	9.25 ± 1.25	40.76 ± 1.21	15.06 ± 1.58	4.13 ± 1.00
	w/o $\ Z \odot Z_c\ _1$	76.66 ± 14.64	16.29 ± 5.32	23.83 ± 1.41	62.66 ± 0.65	30.03 ± 2.57	49.44 ± 0.87	7.91 ± 0.13	41.06 ± 1.76	14.95 ± 1.97	5.18 ± 2.04
	w/o $\ Z - CF\ _F^2$	76.77 ± 14.49	15.30 ± 7.03	23.23 ± 2.13	63.05 ± 0.89	18.93 ± 2.73	50.16 ± 2.82	8.53 ± 3.27	41.16 ± 1.78	13.44 ± 0.38	4.78 ± 0.89
	DREAM	77.74 ± 14.57	19.94 ± 10.87	24.20 ± 1.49	63.63 ± 1.21	33.90 ± 9.49	50.51 ± 2.44	8.99 ± 1.45	41.50 ± 1.55	16.41 ± 2.32	4.77 ± 0.79
NMI	w/o Matrix P	86.36 ± 8.10	16.02 ± 5.24	29.34 ± 2.37	85.15 ± 0.38	45.64 ± 1.56	60.18 ± 0.86	10.82 ± 1.99	51.12 ± 0.55	23.25 ± 1.48	14.62 ± 0.25
	w/o H-enhanced	86.36 ± 8.18	14.79 ± 4.91	29.63 ± 2.00	85.24 ± 0.28	43.26 ± 0.16	60.00 ± 1.30	11.02 ± 1.68	50.93 ± 0.86	21.42 ± 2.45	14.60 ± 0.80
	w/o $\ Z \odot Z_c\ _1$	86.33 ± 8.23	13.31 ± 2.59	30.18 ± 2.00	85.21 ± 0.25	48.15 ± 2.57	59.85 ± 1.46	8.45 ± 0.53	50.95 ± 0.81	23.74 ± 1.36	14.55 ± 0.88
	w/o $\ Z - CF\ _F^2$	86.36 ± 8.18	14.30 ± 5.34	30.01 ± 2.56	85.15 ± 0.16	41.59 ± 2.71	59.33 ± 1.90	10.40 ± 0.39	50.64 ± 1.12	21.34 ± 0.87	13.95 ± 1.54
	DREAM	86.98 ± 8.59	17.81 ± 5.95	30.39 ± 1.31	85.38 ± 0.36	49.00 ± 4.25	60.30 ± 1.40	11.02 ± 1.68	50.95 ± 0.93	23.51 ± 2.68	14.64 ± 0.67
Purity	w/o Matrix P	87.37 ± 9.00	69.59 ± 5.62	42.58 ± 1.45	74.10 ± 1.13	93.95 ± 1.53	68.03 ± 1.97	53.99 ± 1.55	60.39 ± 0.57	49.17 ± 0.97	55.23 ± 2.25
	w/o H-enhanced	87.37 ± 9.03	67.41 ± 5.79	42.67 ± 1.44	74.64 ± 0.66	94.81 ± 0.00	67.59 ± 1.70	52.71 ± 0.00	60.07 ± 1.05	48.35 ± 3.13	54.94 ± 2.96
	w/o $\ Z \odot Z_c\ _1$	87.28 ± 9.16	66.05 ± 4.16	43.44 ± 0.64	73.90 ± 0.80	94.76 ± 0.00	67.83 ± 1.00	52.74 ± 0.12	60.17 ± 1.97	49.16 ± 1.22	54.49 ± 1.39
	w/o $\ Z - CF\ _F^2$	87.38 ± 9.02	67.89 ± 4.95	42.55 ± 1.95	74.62 ± 0.75	92.20 ± 1.61	67.90 ± 2.10	52.31 ± 2.48	59.94 ± 2.27	47.81 ± 0.79	54.95 ± 2.41
	DREAM	88.00 ± 8.99	70.64 ± 4.79	43.51 ± 0.49	75.37 ± 1.26	94.83 ± 1.03	68.79 ± 2.26	52.71 ± 0.00	60.65 ± 1.33	49.61 ± 2.44	55.38 ± 2.52

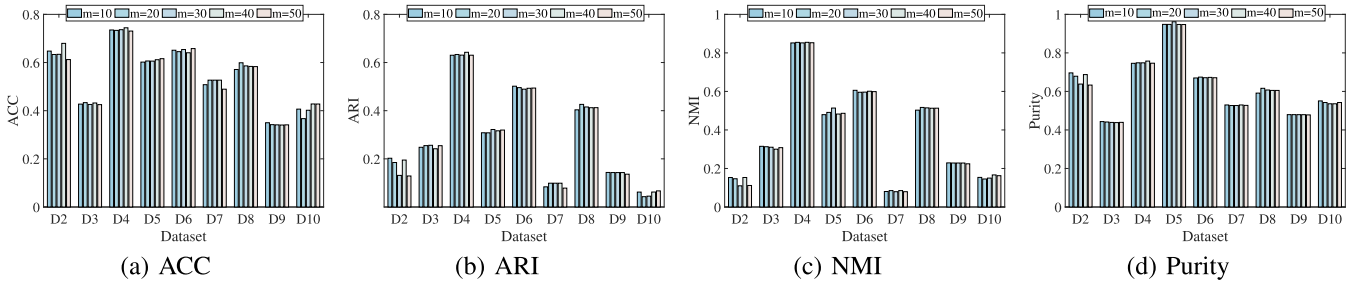


Fig. 8. Experimental results of DREAM w.r.t. different ensemble sizes m , evaluated by four metrics.

summary, these components are overall beneficial to DREAM and jointly contribute to its strong clustering performance.

F. The Influence of Ensemble Size

Fig. 8 shows the performance of the DREAM method with different ensemble sizes m . Although the performance of DREAM does not strictly increase with the number of ensemble members, the smallest ensemble size usually yields the lowest performance, except on a few datasets. The reason why accuracy does not strictly increase with the number of ensemble members is that newly added base clustering results have not been validated and may not necessarily contribute to improving the diversity and accuracy of the ensemble. In ensemble clustering, diversity and accuracy are key factors affecting the overall ensemble performance. While high accuracy positively contributes to the ensemble, low diversity can impair ensemble precision. Currently, few methods that focus on ensemble strategies (consensus functions) take this issue into account. We believe that simultaneously considering the selection of base clustering results and the consensus strategy to enhance ensemble accuracy represents one of the motivations for future developments in ensemble clustering.

Table VIII reports the running time of DREAM under different ensemble sizes. It can be observed that the running time

TABLE VIII
RUNNING TIME COMPARISON UNDER DIFFERENT ENSEMBLE SIZES

Dataset	m=10	m=20	m=30	m=40	m=50
Blobs-Rings	9.5568	13.4980	16.2271	19.4085	23.2093
EMNIST-train	3.5039	5.7072	8.4995	10.0108	12.9244
Winnipeg	2.9350	4.1700	5.4765	6.5803	8.1092
Covtype	5.2483	9.0456	10.1104	13.0096	15.5831

increases as the ensemble size grows on all datasets. This trend is reasonable because a larger ensemble size generally leads to more base clusters, thereby increasing the number of hyperedges involved in hypergraph enhancement. As a result, the costs of random-walk propagation and anchor similarity learning also increase with the ensemble size.

V. CONCLUSION

In this paper, we propose DREAM, a one-step ensemble clustering method based on dynamic anchor learning and hypergraph reconstruction. DREAM first converts the base clusterings into a hypergraph and designs a hypergraph enhancement strategy to improve its reliability. Unlike existing methods, DREAM learns a high-quality anchor similarity matrix through hypergraph reconstruction on an enhanced hypergraph and further refines it

via a local consensus information alignment strategy. DREAM also realizes one-step ensemble clustering by producing a label matrix with a strict clustering interpretation, without requiring any post-processing to obtain the final consensus partition. Extensive experiments on large-scale synthetic and real-world datasets validate the effectiveness of DREAM. On most datasets, DREAM exhibits low sensitivity to its hyperparameters. The dynamic anchor update mechanism incurs additional runtime, so DREAM is not the fastest ensemble method; however, its clustering accuracy is significantly superior to that of existing state-of-the-art methods. In future work, we plan to explore more robust strategies for automatically determining the number of anchors and the similarity threshold, thereby enhancing the practicality of DREAM in fully unsupervised scenarios. We will also investigate streaming or incremental extensions of DREAM, where the enhanced hypergraph and dynamic anchors can be updated more efficiently when new samples or base clustering results arrive.

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